

Python Practice Interview Questions

1. Write a Python Program to Reverse a string taking User input.
2. Write a Python Program to Find out the factorial of a number by taking User input.
3. Check if a string is a palindrome or not.
4. Find the largest element in a list by taking User input list.
5. Write a Python Program to Remove duplicates from a list, lst=[1,2,3,1,2,3,4,5,9,3,1].
6. Write a Python Program to Count occurrences of a character in a string.
7. Write a Python program to generate the Fibonacci sequence up to the nth term, where n is a positive integer.
The program output should be the Fibonacci sequence in a list format.
For Example:
If n = 5, the output should be: [0, 1, 1, 2, 3]
8. Write a Python Program to Check if a number is prime.
9. Write a Python Program to Merge two sorted lists into a single sorted list.
10. Find the common elements between two lists.
11. Find the missing number in a sequence of numbers.
12. Write a Python Program to Find the sum of digits of a number.
13. Write a function to check if a string is an anagram of another string.
14. Convert a string to an integer and handle errors.
15. Find duplicates in a list.
16. Write a Python Program to Count the occurrences of each element in a list.
17. Write a Python Program to Remove all vowels from a string.
18. Write a Python Program to Find the intersection of two lists (common elements).
19. Write a Python Program to Find the nth largest element in a list.
20. Write a method to calculate the power of a number (x^n).
21. Write a Python Program to Reverse the words in a sentence (word-level, not character-level).
22. Write a function to check if a string has all unique characters.
23. Write a function that returns the most frequent element in a list.
24. Write a function to find the longest substring without repeating characters.
25. Implement a basic calculator for addition, subtraction, multiplication, and division.
26. Write a Python Program to Merge two dictionaries in Python.
27. Check if a given list is a valid list of parentheses.
28. Find the second largest number in a list.

29. Write a function to check if a number is a perfect number or not.
30. Write a Python Program to Count the number of vowels in a string.
31. Find the sum of all elements in a list of lists.
32. Check if a string contains all digits.
33. Write a function to remove the punctuation marks from a string.
34. Write a Python Program to Check if a given number is a Armstrong number ($n = \text{sum of its digits raised to the power of the number of digits}$).
35. Write a Python program to perform matrix multiplication.
36. Write a Python Program to Print the Table.
37. Write a Python Program to Print the Right angle triangle and Left angle triangle.
38. Write a Python Program to print the patterns

1.	2.	3.	4.	5.
*	*****	1	1	*
**	*****	12	22	**
***	*****	123	333	***
****	*****	1234	4444	****

39. Difference Between Return and Yield.
40. Difference Between CallByValue and CallByReference.
41. How do you handle large datasets in Python when working with GCP services like BigQuery or Cloud Storage?
42. Write a Python script to read data from a CSV file stored in Google Cloud Storage and load it into BigQuery
43. How would you optimize a Python script that processes data in parallel on Google Cloud Dataflow?
44. Explain how you would use Python to automate the deployment of a data pipeline on GCP (e.g., using Cloud Composer).
45. How do you handle missing or corrupted data in a dataset using Python before loading it into BigQuery?
46. Write a Python function to query data from BigQuery and perform a transformation (e.g., aggregation or filtering).
47. How would you monitor and log errors in a Python-based data pipeline running on GCP?
48. Explain how to use Python to interact with GCP's Pub/Sub for real-time data streaming.
49. How do you ensure data security and compliance when using Python to process data on GCP?
50. Write a Python script to schedule and trigger a Cloud Function that processes data from a Cloud Storage bucket.